

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
M2: Enterprise Java

Practical-8

Roll No.:T017	Name: Radhika mali
Class:TYIT	Batch:1
Date of Assignment: 29-8-2026	Date/Time of Submission: 29-8-2026

Aim:-8. Implement the following EJB applications with different types of Beans

A. Develop simple EJB application to demonstrate Servlet Hit count using Singleton Session Beans.

Code:-

Index.jsp

```
<!DOCTYPE html>
<html>
<head>
  <title>Hit Count Demo</title>
</head>
<body>
<form action="ShowHitCountServlet" method="post">
  <input type="submit"
    value="Take Hit Count demo with EJB Session
Singleton Bean">
</form>
</body>
</html>
```

STHitCountBean.java:-

```
package EJPract;
import javax.ejb.Singleton;
@Singleton
public class STHitCountBean {
  private int count;
  public synchronized int incrHitCount() {
    return count++;
  }
}
```

ShowHitCountServlet.java:-

```
package servlet;
import EJPract.STHitCountBean;
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(name = "ShowHitCountServlet",
  urlPatterns = {"/ShowHitCountServlet"})
public class ShowHitCountServlet extends HttpServlet {
  @EJB
  STHitCountBean cntBean;
```

```
protected void processRequest(HttpServletRequest request,
  HttpServletResponse response)
  throws ServletException, IOException {
  response.setContentType("text/html;charset=UTF-8");
  try (PrintWriter out = response.getWriter()) {
    out.println("<!DOCTYPE html>");
    out.println("<html>");
    out.println("<head>");
    out.println("<title>Servlet
ShowHitCountServlet</title>");
    out.println("</head>");
    out.println("<body>");

    out.println("<h1>You visited this page: "
      + cntBean.incrHitCount()
      + " time/s</h1>");
    out.println("</body>");
    out.println("</html>");
  } }
@Override
protected void doGet(HttpServletRequest request,
  HttpServletResponse response)
  throws ServletException, IOException {
  processRequest(request, response); }
@Override
protected void doPost(HttpServletRequest request,
  HttpServletResponse response)
  throws ServletException, IOException {
  processRequest(request, response); } }
```

Output:-

Take Hit Count demo with EJB Session Singleton Bean

You have visited this page: 1 time/s

B. Develop a simple visitor Statistics application using Message Driven Bean [MDB].**Code:-****Index.jsp**

```

<%@page import="javax.jms.JMSEException"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="javax.jms.TextMessage"%>
<%@page import="javax.jms.MessageProducer"%>
<%@page import="javax.jms.Session"%>
<%@page import="javax.jms.Connection"%>
<%@page import="javax.jms.Queue"%>
<%@page import="javax.jms.ConnectionFactory"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Message Driven Bean</title>
</head>
<body style="background-color:gray;">
<h1>Welcome To Message Driven Bean</h1>
<%
    ConnectionFactory connFact = null;
    Queue queue = null;
    Connection conn = null;
    Session mySess = null;
    MessageProducer msgProducer = null;
    TextMessage msg = null;
    try {
        InitialContext ctx = new InitialContext();
        queue = (Queue) ctx.lookup("jms/Queue");
        connFact = (ConnectionFactory)
            ctx.lookup("jms/QueueFactory");
        conn = connFact.createConnection();
        mySess = conn.createSession(
            false,
            Session.AUTO_ACKNOWLEDGE
        );
        msgProducer = mySess.createProducer(queue);
        msg = mySess.createTextMessage();
        msg.setText(request.getRemoteAddr());
        msgProducer.send(msg);
        conn.close();
    } catch (JMSEException e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    } catch (Exception e) {
        System.out.println(
            "Exception occurred: " + e.toString()
        );
    }
%>
</body>
</html>

```

Sql:-

```

USE reg;
Database changed
CREATE TABLE visitorstat (

```

```

    visitdate TIMESTAMP NOT NULL,
    host VARCHAR(45),
    visitcount INT,
    PRIMARY KEY (host)
);
SELECT * FROM visitorstat;

```

MsggrvnBean.java

```

package EJPract;
import javax.annotation.Resource;
import javax.ejb.ActivationConfigProperty;
import javax.ejb.EJB;
import javax.ejb.MessageDriven;
import javax.ejb.MessageDrivenContext;
import javax.jms.JMSEException;
import javax.jms.Message;
import javax.jms.MessageListener;
import javax.jms.TextMessage;
@MessageDriven(
    activationConfig = {
        @ActivationConfigProperty(
            propertyName = "destinationType",
            propertyValue = "javax.jms.Queue"
        ),
        @ActivationConfigProperty(
            propertyName = "destinationLookup",
            propertyValue = "jms/Queue"
        )
    }
)
public class MsgDrvnBean implements MessageListener {
    @EJB
    VStatelessBean vs;
    @Resource
    private MessageDrivenContext mdc;
    public MsgDrvnBean() { }
    @Override
    public void onMessage(Message message) {
        try {
            if (message instanceof TextMessage) {
                TextMessage msg =
                    (TextMessage) message;
                vs.addVisitor(msg.getText());
            }
            catch (JMSEException e) {
                mdc.setRollbackOnly();
            }
        }
    }
}

```

VStatelessBean.java

```

package EJPract;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import javax.annotation.PostConstruct;
import javax.annotation.PreDestroy;
import javax.ejb.Stateless;
@Stateless
public class VStatelessBean {
    private Connection conn = null;
    private ResultSet rs;

```

```

private Statement stmt = null;
private String query = null;
@PostConstruct
public void connect() {
    try {
        Class.forName("com.mysql.jdbc.Driver");
        String connStr =
            "jdbc:mysql://localhost/reg";
        conn = DriverManager.getConnection(
            connStr,
            "root",
            "root"
        );
        System.out.println("Connection Success.");
    } catch (ClassNotFoundException | SQLException e) {
        System.err.println("Connection failed: " +
            e.getMessage());
    } }
@PreDestroy
public void disconnect() {
    try {
        conn.close();
        System.out.println(
            "Connection closed."
        );
    } catch (SQLException e) {
        System.err.println(
            "Not able to close connection: "
            + e.getMessage()
        );
    } }
public void addVisitor(String host) {
    try {
        stmt = conn.createStatement();

```

```

query =
    "INSERT INTO visitorstat(host,visitcount) " +
    "VALUES('" + host + "',1)";

stmt.executeUpdate(query);
    } catch (SQLException e) {
        try {
            stmt = conn.createStatement();
            query =
                "UPDATE visitorstat " +
                "SET visitcount = visitcount + 1 " +
                "WHERE host = '" + host + "'";
            stmt.executeUpdate(query);
        } catch (SQLException ex) {
            System.err.println(
                "Cannot update: "
                + ex.getMessage()
            );
        }
    } } }

```

Output:-**Welcome To Message Driven Bean**

```

mysql> USE reg;
Database changed
mysql> SELECT * FROM visitorstat;
+-----+-----+-----+
| visitdate | host | visitcount |
+-----+-----+-----+
| 2026-08-29 11:24:28 | 0:0:0:0:0:0:1 | 18 |
+-----+-----+-----+
1 row in set (0.00 sec)

```

C. Develop simple Marks Entry Application to demonstrate accessing Database using EJB.**Code:****Index.jsp**

```

<%@page import="java.util.ArrayList"%>
<%@page import="java.util.Iterator"%>
<%@page import="java.util.List"%>
<%@page import="javax.naming.InitialContext"%>
<%@page import="EJPract.MarksEntryBean"%>
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<%!
private static MarksEntryBean marksEntry;
public void jspInit() {
    try {
        InitialContext ic = new InitialContext();
        marksEntry = (MarksEntryBean)
            ic.lookup(
                "java:global/EJBMarksEntry/MarksEntryBean"
            );
    } catch (Exception ex) {
        System.out.println(
            "Database Error " + ex.getMessage()
        );
    } } %>
<%
if (request.getParameter("insertMarks") != null) {

```

```

String name, marks1, marks2, marks3;
name = request.getParameter("stuName");
marks1 = request.getParameter("sub1");
marks2 = request.getParameter("sub2");
marks3 = request.getParameter("sub3");
marksEntry.addMarks(
    name,
    marks1,
    marks2,
    marks3
); } %>
<html>
<head>
    <meta http-equiv="Content-Type"
        content="text/html; charset=UTF-8">
    <title>Marks Entry</title>
</head>
<body>
<h2 style="text-align:center;">
    Enter Details
</h2> <br>
<form name="result" method="post">
<table>
<tr>

```

Radhika 17

```
<td>Enter Students Name</td>
<td>
  <input type="text"
    name="stuName"
    value="">
</td> </tr>
<tr>
<td>Enter Subject 1 Marks</td>
<td>
  <input type="text"
    name="sub1"
    value="">
</td> </tr>
<tr>
<td>Enter Subject 2 Marks</td>
<td>
  <input type="text"
    name="sub2"
    value="">
</td> </tr>
<tr>
<td>Enter Subject 3 Marks</td>
<td>
  <input type="text"
    name="sub3"
    value="">
</td> </tr>
<tr>
<td colspan="2">
  <input type="submit"
    value="insertMarks"
    name="insertMarks">
</td> </tr>
</table> </form> </body> </html>
```

MarksEntryBean.java:-

```
package EJPract;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;
import javax.ejb.Stateful;
@Stateful
```

```
public class MarksEntryBean {
    List<String> details;
    String stuName, sub1, sub2, sub3;
    private Connection conn = null;
    private ResultSet rs;
    private Statement stmt = null;
    private String query = null;
    public void createConnection() {
        try {
            Class.forName("org.gjt.mm.mysql.Driver");
            String connStr =
                "jdbc:mysql://localhost/reg";
            conn = DriverManager.getConnection(
                connStr,
                "root",
                "root"
            );
        } catch (ClassNotFoundException | SQLException e) {
            System.err.println(
                "Database Error " + e.getMessage()
            );
        }
    }
    public void addMarks(
        String stuname,
        String marks1,
        String marks2,
        String marks3) {
        createConnection();
        try {
            stmt = conn.createStatement();
            query =
                "INSERT INTO marks" +
                "(stuName,sub1,sub2,sub3)" +
                "VALUES('" +
                stuname + "','" +
                marks1 + "','" +
                marks2 + "','" +
                marks3 + "')";
            stmt.executeUpdate(query);
        } catch (SQLException e) {
            System.err.println(
                "Failed to insert. "
                + e.getMessage()
            );
        }
    }
}
```

Output:

mysql> SELECT * FROM marks;				
id	stuName	sub1	sub2	sub3
1	aashu	100	69	100
2	Asmitha	100	100	100
3	Afsha	100	100	100
4	Radhika	1000	100	100
4 rows in set (0.00 sec)				